FACTORING A MORPHISM THROUGH A CLOSED SUBSPACE

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Editorial note. Counted span is the author's own proof environment, source lines 642-670, following the statement at 631-640.

Editorial note. The omitted verification (~666-668) is the stalk-locality check for the assembled pair (g, g^sharp): the composite O_{X,f(y)} -> O_{Z,g(y)} -> O_{Y,y} equals the local map f^sharp_y and the first arrow is a surjective local quotient map, forcing g^sharp_y local; mechanical, same precedent as cards L26348/L26363.

Editorial note. Uniqueness is already supplied by the named Sheaves engine (~664-665, ~669).

Allowed standard prerequisites (editorial).

- Closed subspace from a sheaf of ideals locally generated by sections. definition-closed-subspace (same chapter, statement at 606). Z = Supp(O_X/I) with structure sheaf pushed forward from O_X/I.
- Closed immersions and sheaf category equivalence. Sheaves, Lemma sheaves-lemma-equivalence-categories-closed-structures (cross-chapter, cited at ~663). Produces the unique g^sharp from the quotient sheaf map.
- Adjointness of f_* and f^*. Standard sheaf adjunction (used at ~656-658). Converts vanishing of f^*I -> O_Y into vanishing of I -> f_*O_Y.

Source reference. Original reference 00AH: focused-seeds/geometry/S9-schemes-characterize-closed-subspace/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X, Y be locally ringed spaces. Let $\mathcal{I} \subset \mathcal{O}_X$ be a sheaf of ideals locally generated by sections. Let $i : Z \to X$ be the associated closed subspace. A morphism $f : Y \to X$ factors through Z if and only if the map $f^*\mathcal{I} \to f^*\mathcal{O}_X = \mathcal{O}_Y$ is zero. If this is the case the morphism $g : Y \to Z$ such that $f = i \circ g$ is unique.*

Proof. Clearly if f factors as $Y \to Z \to X$ then the map $f^*\mathcal{I} \to \mathcal{O}_Y$ is zero. Conversely suppose that $f^*\mathcal{I} \to \mathcal{O}_Y$ is zero. Pick any $y \in Y$, and consider the ring map $f_y^\sharp : \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$. Since the composition $\mathcal{I}_{f(y)} \to \mathcal{O}_{X,f(y)} \to \mathcal{O}_{Y,y}$ is zero by assumption and since $f_y^\sharp(1) = 1$ we see that $1 \notin \mathcal{I}_{f(y)}$, i.e., $\mathcal{I}_{f(y)} \neq \mathcal{O}_{X,f(y)}$. We conclude that $f(Y) \subset Z = \mathrm{Supp}(\mathcal{O}_X/\mathcal{I})$. Hence $f = i \circ g$ where $g : Y \to Z$ is continuous. Consider the map $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$. The assumption $f^*\mathcal{I} \to \mathcal{O}_Y$ is zero implies that the composition $\mathcal{I} \to \mathcal{O}_X \to f_*\mathcal{O}_Y$ is zero by adjointness of f_* and f^*. In other words, we obtain a morphism of sheaves of rings $\overline{f^\sharp} : \mathcal{O}_X/\mathcal{I} \to f_*\mathcal{O}_Y$. Note that $f_*\mathcal{O}_Y = i_*g_*\mathcal{O}_Y$ and that $\mathcal{O}_X/\mathcal{I} = i_*\mathcal{O}_Z$. By Sheaves, Lemma 00AH we

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obtain a unique morphism of sheaves of rings $g^\sharp : \mathcal{O}_Z \to g_*\mathcal{O}_Y$ whose pushforward under i is $\overline{f^\sharp}$. We omit the verification that $(g, g^\sharp)$ defines a morphism of locally ringed spaces and that $f = i \circ g$ as a morphism of locally ringed spaces. The uniqueness of $(g, g^\sharp)$ was pointed out above. $\square$